



DIGITAL TECHNOLOGY

Academic Year 2024-25

Prof. Barbara PERNICI

Exam 26/1/2026 - B

Total time 1.30 h

Last name

First name

Matricola/Person code

Signature

Please remember that:

- The exam is closed books.
- The use of cellular phones or any other electronic devices during the exam is forbidden.

It is necessary to answer at least partially all the questions for a positive evaluation.

Given answers should be explained, lists of bullet items are insufficient to answer a question.

Please write the answer for each question on a separate piece of paper.

Question 1 [11 points]

Illustrate the creation of a digital signature and illustrate in which applications digital signatures can be used.

Question 2 [11 points]

Define the main data quality dimensions analyzed in the course.

[please turn over]

Question 3 [11 points]

Consider the following fragment of Python code:

```
cities = ["Rome", "Paris", "Rome", "Berlin", "Paris"]
temps = [20, 18, 22, 16, 19]

pairs = []
for i in range(len(cities)):
    pairs.append((cities[i], temps[i]))

def datafilter(data, limit):
    result = []
    for city, temp in data:
        if temp >= limit:
            result.append((city, temp))
    return result

filtered = datafilter(pairs, 19)
```

Questions:

1. What is the content of the variable `filtered` after running all the code?
2. Rewrite the function `datafilter` so that it returns a dictionary.
3. Using a Pandas DataFrame, compute the average temperature and subtract it from all the records.